

2022
Higher School Certificate
Trial Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student name and/or number at the top of every page

Total marks – 100

Section I – 10 marks (pages 3 - 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 - 22)

- Attempt Questions 11 – 33
- Allow about 2 hours and 45 minutes for this section

This paper MUST NOT be removed from the examination room.

STUDENT NAME/NUMBER.....

STUDENT NAME/NUMBER.....

Section I

10 Marks

Attempt Questions 1-10.

Allow about 15 minutes for this section.

Select the alternative A, B, C, D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
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8				
9				
10				

Section I**10 Marks****Attempt Questions 1-10.****Allow about 15 minutes for this section.****Use the multiple-choice answer sheet for questions 1-10.**

1. A sector with perimeter 28 cm contains an angle of 1.5 radians at the centre of a circle with radius r cm. What is the value of r ?
(A) 6
(B) 8
(C) 10
(D) 12

2. Consider the functions $f(x) = x^2$ and $g(x) = x + 2$. Which of the following expressions is equal to $f(g(x)) - g(f(x))$?
(A) $4x + 2$
(B) $4x + 4$
(C) $x^2 - x - 2$
(D) 2

3. A and B are two independent events with $P(A \cup B) = \frac{3}{8}$ and $P(A) = \frac{1}{4}$.
What is the value of $P(B)$?
(A) $\frac{1}{12}$
(B) $\frac{1}{10}$
(C) $\frac{1}{8}$
(D) $\frac{1}{6}$

4. What is the domain of the function $f(x) = \ln(x-1) + \sqrt{2-x}$?
(A) $(1, 2)$
(B) $(1, 2]$
(C) $[1, 2)$
(D) $[1, 2]$

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5. The function $f(x) = e^x$ is transformed into the function $g(x) = e^{x+2}$. Which of the following describes the effect of this transformation?

- (A) a horizontal dilation
- (B) a vertical dilation
- (C) a vertical translation
- (D) none of the above

6. Which of the following is an expression for $\frac{d}{dx} \log_{10} 3x$?

- (A) $\frac{1}{3x \log_{10} e}$
- (B) $\frac{1}{x \log_{10} e}$
- (C) $\frac{\log_{10} e}{3x}$
- (D) $\frac{\log_{10} e}{x}$

7. The marks on a test are normally distributed with mean μ and standard deviation σ .

2.5% of the marks are 34 or less and 16% of the marks are 70 or more. What is the value of μ ?

- (A) 54
- (B) 56
- (C) 58
- (D) 60

8. Which of the following is an expression for $\int \left(\frac{1}{1+\sin x} + \frac{1}{1-\sin x} \right) dx$?

- (A) $2 \sec^2 x + c$
- (B) $4 \tan x \sec^2 x + c$
- (C) $2 \tan x + c$
- (D) $2x + 2 \sec x + c$

9. A single lump sum investment of $\$P$ earns an amount of interest $\$I$ after 2 years with a rate of interest of 6% p.a. compounded monthly. Which of the following is an expression for P ?

(A) $\frac{I}{(1.06)^2 + 1}$

(B) $\frac{I}{(1.06)^2 - 1}$

(C) $\frac{I}{(1.005)^{24} + 1}$

(D) $\frac{I}{(1.005)^{24} - 1}$

10. If $x > 0$ what is the limiting sum of the series

$$\log_2 x - \log_4 x + \log_{16} x - \log_{256} x + \dots + (-1)^{n+1} \log_{2^{(2^n)}} x + \dots ?$$

(A) $\frac{2}{3} \log_2 x$

(B) $\frac{3}{4} \log_2 x$

(C) $\frac{4}{5} \log_2 x$

(D) $\frac{5}{6} \log_2 x$

Section II

90 marks

Attempt Questions 11 – 33

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. If extra space is required, use the lined pages at the end of this section, carefully labelling the additional work with the appropriate question number and part.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

A circle has equation $x^2 + y^2 - 6x + 4y + k = 0$ for some constant k . Find the set of possible values of k .

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Question 12 (2 marks)

If $\tan x = 3$ and $\sin x < 0$ find the exact value of $\cos x$.

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Question 13 (2 marks)

A discrete random variable X with mean 4 has probability distribution

2

x	a	$2a$	$3a$	$4a$
$p(x)$	$4p$	$3p$	$2p$	p

for some constants a and p . Find the values of a and p .

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Question 14 (3 marks)

$A(-1, -5)$ and $B(5, -1)$ are two points. Find in general form the equation of the straight line through the midpoint of AB and perpendicular to AB .

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Question 15 (3 marks)

Find the coordinates of the point on the curve $y = 2\sqrt{x}$ where the tangent to the curve is parallel to the line $2x - y - 2 = 0$.

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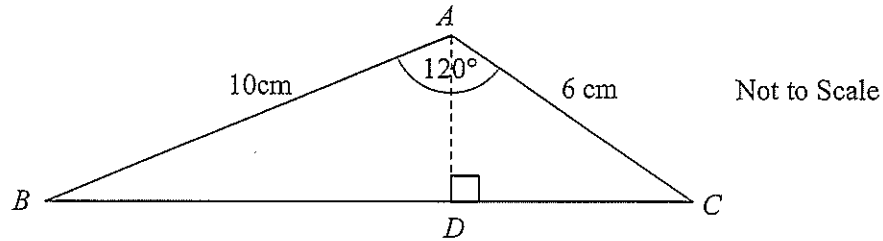
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Question 16 (4 marks)

In the diagram, ABC is a triangle in which $AB = 10\text{ cm}$, $AC = 6\text{ cm}$ and $\angle CAB = 120^\circ$. AD is the perpendicular from A to BC .

(a) Find the length of BC

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(b) Find the length of AD in centimetres correct to one decimal place.

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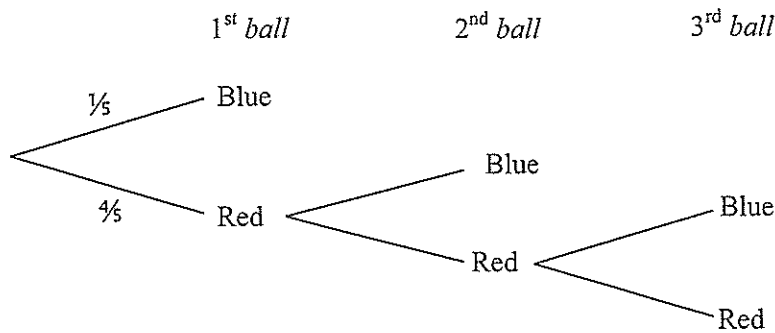
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Question 17 (4 marks)

A bag contains 1 blue and 4 red balls. A first ball is chosen at random without replacement from the bag. If this first ball is blue, no further balls are chosen. If the first ball is red, a second ball is chosen at random without replacement from the remaining balls. If this second ball is blue, no further balls are chosen. If the second ball is red, a third ball is chosen at random without replacement from the remaining balls.

(a) Complete the probability tree diagram below:

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(b) Find the probability that 3 red balls are chosen.

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(c) Find the probability that at least one ball of each colour is chosen.

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Question 18 (2 marks)

Given that $\ln a - \ln b = \ln(a - b)$ where $a > b > 1$ show that $a = \frac{b^2}{b-1}$.

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Question 19 (2 marks)

Find correct to 2 decimal places the gradient of the curve $y = x \sin x$ where $x = 1$. 2

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Question 20 (2 marks)

Show that the curve $y = x^2 - 8 \ln x$ is concave up for all values of $x > 0$. 2

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Question 21 (4 marks)

At the Oztown school, HSC Trial marks in English Standard had a mean of 72 and a standard deviation of 8, whilst HSC Trial marks in Mathematics Standard 2 had a mean of 68 and a standard deviation of 10.

(a) Otto scored 66 in the English Standard Trial and 63 in the Mathematics Standard 2 Trial. 2
Explain which of these was Otto's better performance.

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(b) Anna scored 84 in the English Standard Trial but sat the Mathematics Standard 2 Trial late. 2
Find what mark Anna needed to score in the Mathematics Standard 2 Trial so that she performed equally well in both subjects.

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Question 22 (3 marks)

Find the values of any constant A such that $\int_0^3 (x^2 + A) dx = 2A^2$.

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Question 23 (3 marks)

Find the value of the constant $k > 0$ such that $\int_{\ln k}^{\ln k^2} \left(1 + \frac{1}{x}\right) dx = \ln 10$.

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Marks

Find the coordinates and nature of the stationary points on the curve $y = \frac{x^2}{x-2}$.

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Question 25 (4 marks)

The table below shows the future values of an annuity of \$1 for different rates of interest and for different numbers of compounding periods. The contributions are made at the end of each time period.

Future Value Interest Factors

<i>Time Period</i>	<i>Interest Rate</i>				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

- (a) An annuity account is to be opened in order to save \$5000 in 2 years time. Equal instalments are to be made at the end of each quarter for 2 years at an interest rate of 12% p.a. compounding quarterly. Find the amount of each quarterly instalment.

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- (b) Find the amount in the annuity account after 1 year.

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STUDENT NAME/NUMBER.....

Marks

Question 26 (4 marks)

At time t hours after midnight on a particular day the depth D metres of water in the Oztown harbour is given by the function $D = 8 + 2\cos\left(\frac{4\pi}{25}t\right)$ for $0 \leq t \leq 24$.

- (a) Find the amplitude and period of the function.

2

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- (b) Sketch the graph of D as a function of t clearly showing the times that day when the water level was at its lowest and highest.

2

STUDENT NAME/NUMBER.....

Marks

Question 27 (4 marks)

An item is purchased for \$20 000. It appreciates in value so that after t years its value $\$V$ is given by the formula $V = 20000p^t$ for some constant $p > 1$. After 9 years its value is \$32 000. 2

(a) Find the value of p correct to 4 decimal places.

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(b) Find correct to the nearest \$10 per year the rate at which the item is appreciating in value after 9 years. 2

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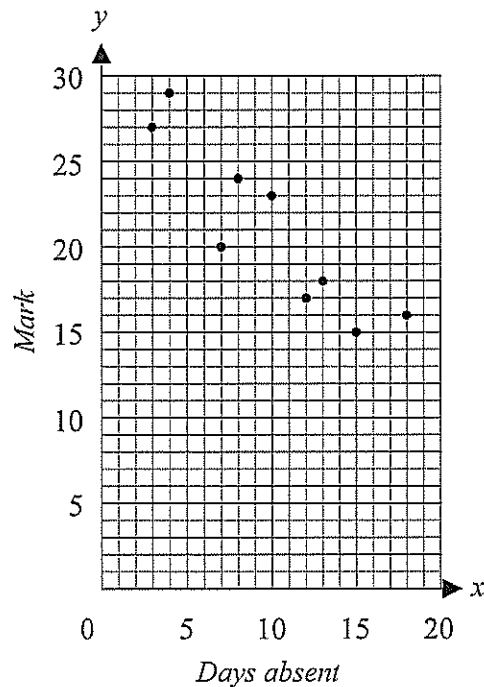
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Question 28 (5 marks)

The table and scatter plot below show the end of term class test marks and the number of days absent up until the day of the class test for 9 of the 10 students in a class. Student J missed the end of term test but had 11 days absent up until the day of the class test.

Student	A	B	C	D	E	F	G	H	I
Days Absent (x)	3	4	7	8	10	12	13	15	18
Mark (y)	27	29	20	24	23	17	18	15	16



- (a) Use the scatter plot to describe the nature and strength of the linear relationship between *Days Absent* and *Marks*. 1

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- (b) On the scatter plot, draw the line of best fit to pass through the points $(0, 30)$ and $(10, 21)$. 3
Find the equation of this line of best fit.

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- (c) Estimate correct to the nearest whole number the test mark for student J. 1

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At the start of year 2000 a colony of 600 animals was established in a sanctuary. Each year the population size grows at a rate of 10% per year and at the end of each year 50 of the animals are released into the wild. The population size P_n at the end of the n^{th} year can be determined using the recurrence relation $P_n = 1.1 \times P_{n-1} - 50$ where $n = 1, 2, 3, \dots$ and $P_0 = 600$.

- 3

$$P_n = 100 \times (1.1)^n + 500.$$

- 2

Question 30 (6 marks)

A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line and velocity $v = 3\sqrt{t+1} - 9$ metres per second. Initially the particle is at O .

(a) Use integration to show that $x = 2(t+1)\sqrt{t+1} - 9t - 2$. **2**

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(b) Find when and where the particle is at rest. **2**

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(c) Find the distance travelled by the particle in the first 24 seconds of its motion. **2**

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Question 31 (7 marks)

The cost $\$C$ per hour when employees work x hours per day to complete a project is given by $C = ax^3 + d$ for some constants a and d . The time T hours to complete the project is given by $T = \frac{120}{x}$. In order to complete the project, the cost would be $\$248$ per hour when employees work 4 hours per day on the project and $\$472$ per hour when employees work 8 hours per day on the project.

- (a) Write down two equations in a and d and solve these equations to find the values of a and d . 3

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- (b) Show that the total cost $\$S$ of completing the project when employees work x hours per day on the project is given by $S = 60x^2 + \frac{25\,920}{x}$. 1

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- (c) Use differentiation to find the minimum total cost of completing the project. 3

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Question 32 (7 marks)

A continuous random variable X has probability density function $f(x)$ given by

$$f(x) = \frac{2}{x^2} \text{ for } 1 \leq x \leq 2 \quad \text{and} \quad f(x) = 0 \text{ for } x < 1 \text{ or } x > 2.$$

(a) Show that $f(x)$ is a probability density function.

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(b) Find the cumulative distribution function $F(x)$ of the random variable X .

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(c) Show that if Q_1, Q_2, Q_3 are the quartiles of the distribution of X then $\frac{1}{Q_1}, \frac{1}{Q_2}, \frac{1}{Q_3}$ are consecutive terms in an arithmetic sequence.

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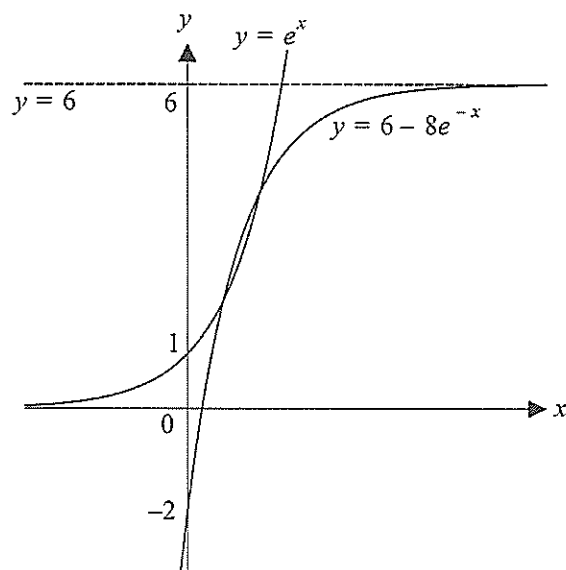
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Question 33 (8 marks)



The diagram shows the graphs of the curves $y = f(x)$ and $y = g(x)$ where $f(x) = e^x$ and $g(x) = 6 - 8e^{-x}$.

- (a) State a sequence of dilations, reflections, translations that would transform the graph of the curve $y = f(x)$ into the graph of the curve $y = g(x)$.

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- (b) Show that the x coordinates (x_1 and x_2 where $x_1 < x_2$) of the points of intersection of the two curves are the solutions of the equation $e^{2x} - 6e^x + 8 = 0$ and solve this equation to find the exact values of x_1 and x_2 .

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Question 33 is continued on the next page

STUDENT NAME/NUMBER.....

Marks

Question 33 continued.

(c) Find in simplest exact form the area of the region enclosed by the two curves.

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Section II Extra writing space.

If you use this space, clearly indicate which question you are answering.

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STUDENT NAME/NUMBER.....

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Section 1 Questions 1-10 (1 mark each)

Question	Answer	Solution	Outcomes
1	B	$r \times 1.5 + 2r = 28 \quad \frac{7}{2}r = 28 \quad \therefore r = 8$	MA11-3
2	A	$f(g(x)) - g(f(x)) = (x+2)^2 - (x^2+2) = 4x+2$	MA11-2
3	D	$P(A \cup B) = P(A) + P(B) - P(A)P(B)$ since A, B are independent $\frac{3}{8} = \frac{1}{4} + P(B)(1 - \frac{1}{4}) \quad \therefore P(B) = \frac{1}{8} \times \frac{4}{3} = \frac{1}{6}$	MA11-7
4	B	$\ln(x-1)$ has domain $x > 1$ and $\sqrt{2-x}$ has domain $x \leq 2$ hence $f(x)$ has domain $1 < x \leq 2$. In interval notation, $(1, 2]$	MA11-2
5	B	$g(x) = e^{x+2} = e^2 e^x$ Hence a vertical dilation by a factor e^2	MA12-1
6	D	$\frac{d}{dx} \log_{10} 3x = \frac{d}{dx} \left(\log_{10} 3 + \frac{\ln x}{\ln 10} \right) = \frac{1}{x \ln 10} = \frac{\log_{10} e}{x \log_{10} 10} = \frac{\log_{10} e}{x}$	MA12-6
7	C	$\mu - 2\sigma = 34$ $\mu + \sigma = 70 \quad \therefore 3\mu = 34 + 2 \times 70 = 174 \quad \therefore \mu = 58$	MA12-8
8	C	$\int \left(\frac{1}{1+\sin x} + \frac{1}{1-\sin x} \right) dx = \int \frac{2}{1-\sin^2 x} dx = \int \frac{2}{\cos^2 x} dx = 2 \int \sec^2 x dx$ Hence expression is $2 \tan x + c$	MA12-7
9	D	$P(1 + \frac{0.005}{12})^{24} - P = I \quad P\{(1.005)^{24} - 1\} = I \quad \therefore P = \frac{I}{(1.005)^{24} - 1}$	MA12-2
10	A	$T_n = (-1)^{n+1} \frac{\log_2 x}{\log_2 2^{(2^{n-1})}} = (-1)^{n+1} \frac{\log_2 x}{2^{n-1}}$ Hence series is geometric with $a = \log_2 x$ and $r = -\frac{1}{2}$. Limiting sum is $\frac{\log_2 x}{1 - (-\frac{1}{2})} = \frac{2}{3} \log_2 x$	MA12-4

Section 2

Question 11

Outcomes assessed: MA11-2

Marking Guidelines

Criteria	Marks
Completes the square to express r^2 in terms of k to deduce that $k < 13$	2
Substantial progress eg. correct procedure with one error in algebra or quotes $k \leq 13$	1

Answer

$$\begin{aligned}
 x^2 + y^2 - 6x + 4y + k &= 0 \\
 x^2 - 6x + 9 + y^2 + 4y + 4 &= 13 - k & \text{Hence } r^2 = 13 - k \quad \therefore k < 13 \\
 (x-3)^2 + (y+2)^2 &= 13 - k
 \end{aligned}$$

Question 12

Outcomes assessed: MA11-3

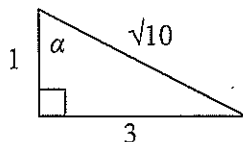
Marking Guidelines

Criteria	Marks
Identifies x as a 3 rd quadrant angle then uses a trig identity or Pythagoras to find $\cos x$	2
Substantial progress eg finds the correct numerical value but with the wrong sign	1

Answer

x is a 3rd quadrant angle.

Let α be the related acute angle



$$\cos \alpha = \frac{1}{\sqrt{10}}$$

$$\cos x = -\frac{1}{\sqrt{10}}$$

Question 13

Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Uses the values of the mean and probability sum to find the values of p, a	2
Substantial progress eg. writes one equation for a, p using the mean or probability sum	1

Answer

$$\sum p(x) = 1 \Rightarrow 10p = 1 \quad \therefore p = \frac{1}{10}$$

$$E(X) = 4 \Rightarrow 4ap + 6ap + 6ap + 4ap = 4 \quad \therefore 20ap = 4 \quad \therefore a = 2$$

Question 14

Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Finds the midpoint and gradient of AB to find the required equation	3
Substantial progress eg. correct procedure but with one error	2
Some progress eg, finds the midpoint or gradient of AB	1

Answer

Midpoint of AB has coordinates $(2, -3)$ and gradient of AB is $\frac{2}{3}$

$$\perp \text{ bisector of } AB \text{ has equation } y + 3 = -\frac{3}{2}(x - 2)$$

$$2y + 6 = -3x + 6$$

Hence required line has equation

$$3x + 2y = 0$$

Question 15

Outcomes assessed: MA11-5

Marking Guidelines

Criteria	Marks
Uses differentiation to find x where tangent and line have same gradient then finds y	3
Substantial progress eg correct procedure but with one error	2
Some progress eg differentiates correctly	1

Answer

$$\begin{aligned}
 2x - y - 2 &= 0 & y &= 2\sqrt{x} & \frac{dy}{dx} &= 2 \Rightarrow \sqrt{x} = \frac{1}{2} \\
 y &= 2x - 2 & \frac{dy}{dx} &= \frac{1}{\sqrt{x}} & \therefore \text{point has coordinates } & \left(\frac{1}{4}, 1\right) \\
 \text{Line has gradient } &2.
 \end{aligned}$$

Question 16

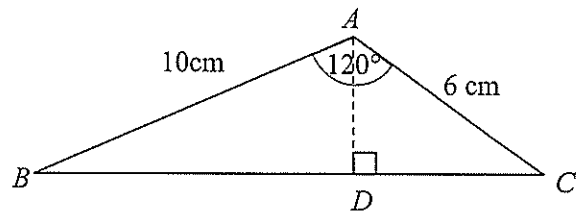
a. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Applies the cosine rule to find the length of BC	2
Substantial progress eg writes a numerical expression for BC^2 using the cosine rule	1

Answer

$$\begin{aligned}
 BC^2 &= 10^2 + 6^2 - 120 \cos 120^\circ = 196 \\
 \therefore BC &= 14 \text{ cm}
 \end{aligned}$$



b. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Applies appropriate trigonometric results to deduce the length of AD	2
Substantial progress eg finds $\sin B$, $\sin C$, $\cos B$, $\cos C$ or Area $\triangle ABC$	1

Answer

$$\text{Area } \triangle ABC = \frac{1}{2} \times 10 \times 6 \sin 120^\circ = 15\sqrt{3} \quad \therefore \frac{1}{2} \times BC \times AD = 15\sqrt{3} \quad \therefore AD \approx 3.7 \text{ cm (1 dec. pl)}$$

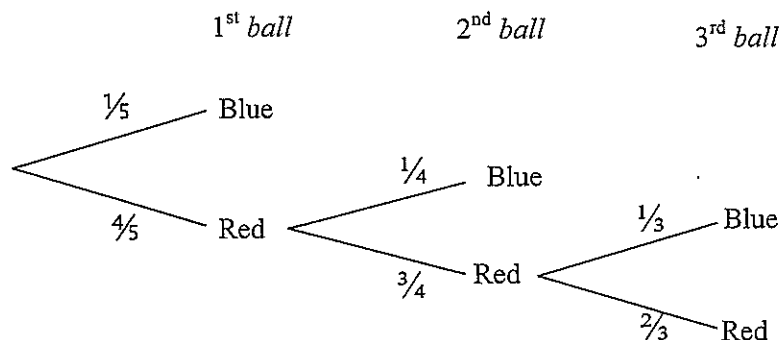
Question 17

a. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Completes probability tree	1

Answer



b. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Correct answer	1

Answer

$$P(R, R, R) = \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} = \frac{2}{5}$$

c. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Calculates required probability	2
Substantial progress eg identifies one appropriate path through the tree with its probability	1

Answer

$$P(R, B) + P(R, R, B) = \frac{4}{5} \times \frac{1}{4} + \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{2}{5}$$

Question 18

Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Applies log laws to write a relation for a and b then writes a in terms of b	2
Substantial progress eg writes a relation for a and b using log laws	1

Answer

$$\ln a - \ln b = \ln(a - b) \Rightarrow \frac{a}{b} = a - b \quad \text{Then} \quad \begin{aligned} a &= ab - b^2 \\ b^2 &= a(b - 1) \end{aligned} \quad \therefore a = \frac{b^2}{b - 1}$$

Question 19

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Finds the value of the first derivative when $x=1$	2
Substantial progress eg finds the first derivative	1

Answer

$$y = x \sin x$$

$$\frac{dy}{dx} = \sin x + x \cos x$$

$$x = 1 \Rightarrow \frac{dy}{dx} \approx 1.38$$

$\therefore$ gradient of tangent is 1.38

Question 20

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Shows second derivative is positive for all $x > 0$	2
Substantial progress eg finds the first derivative	1

Answer

$$y = x^2 - 8 \ln x$$

$$\frac{dy}{dx} = 2x - \frac{8}{x}$$

$$\frac{d^2y}{dx^2} = 2 + \frac{8}{x^2}$$

$$\therefore \frac{d^2y}{dx^2} > 0 \text{ for all } x > 0$$

$\therefore$ curve is concave up for all $x > 0$

Question 21

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Compares z-scores to make correct decision	2
Substantial progress eg calculates one z-score	1

Answer

$$z_{\text{English}} = \frac{66-72}{8} = -0.75$$

$$z_{\text{Maths}} = \frac{63-68}{10} = -0.5$$

Performance in Maths was better.

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Writes and solves an equation for Maths mark x that would give same z-score as that for English	2
Substantial progress eg writes an equation for x	1

Answer

For equal z-scores, $\frac{x-68}{10} = \frac{84-72}{8}$ $\therefore x-68=15$ Anna needs to score 83 in the Maths exam.

Question 22

Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Integrates to find and solve a quadratic equation for A	3
Substantial progress eg integrates to find a quadratic equation for A	2
Some progress eg finds the definite integral in terms of A	1

Answer

$$\int_0^3 (x^2 + A) dx = \left[\frac{1}{3}x^3 + Ax \right]_0^3 = 9 + 3A$$

$$9 + 3A = 2A^2$$

$$2A^2 - 3A - 9 = 0$$

$$(2A + 3)(A - 3) = 0$$

$$\therefore A = 3 \text{ or } A = -\frac{3}{2}$$

Question 23

Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Evaluates definite integral in terms of k then writes and solves equation for k	3
Substantial progress eg evaluates and partially simplifies definite integral in terms of k	2
Some progress eg evaluates definite integral in terms of k without simplification	1

Answer

$$\int_{\ln k}^{\ln k^2} \left(1 + \frac{1}{x} \right) dx = \left[x + \ln x \right]_{\ln k}^{\ln k^2}$$

$$= (\ln k^2 - \ln k) + \{ \ln(\ln k^2) - \ln(\ln k) \}$$

$$= 2 \ln k - \ln k + \ln(2 \ln k) - \ln(\ln k)$$

$$= \ln k + \ln 2 + \ln(\ln k) - \ln(\ln k)$$

$$= \ln 2k$$

$$\therefore \ln 2k = \ln 10$$

$$k = 5$$

Question 24

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Differentiates to find both stationary points and determines their nature	4
Substantial progress eg finds the stationary points and the nature of one of them	3
Moderate progress eg finds the stationary points	2
Some progress eg applies the quotient rule for differentiation	1

Answer

$$y = \frac{x^2}{x-2}$$

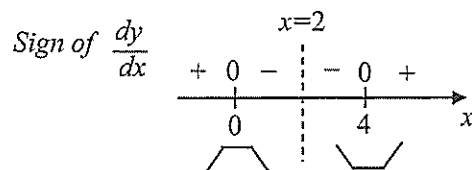
$$\frac{dy}{dx} = \frac{2x(x-2) - x^2}{(x-2)^2}$$

$$= \frac{x^2 - 4x}{(x-2)^2}$$

$$\frac{dy}{dx} = \frac{x(x-4)}{(x-2)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow \begin{cases} x=0, y=0 \\ \text{or} \\ x=4, y=8 \end{cases}$$

Stationary points are (0, 0) and (4, 8)



(0, 0) is a maximum turning point. (4, 8) is a minimum turning point

Question 25

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Uses appropriate table entry to calculate the quarterly instalment	2
Substantial progress eg correct table value selected, or correct procedure but error in table value	1

Answer

Future value of \$1 annuity for 8 quarters at 3% per quarter is \$8.8923.

Quarterly instalment \$a is given by $8.8923 \times a = 5000$

$$a = 5000 \div 8.8923$$

Ans. \$562.28 quarterly

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Uses appropriate table entry and answer from (a) to calculate the future value after 1 year	2
Substantial progress eg selects correct table value	1

Answer

Future value of annuity of \$562.28 quarterly for 4 quarters is $4.1836 \times \$562.28 = \2352.35

Question 26

a. Outcomes assessed: MA12-5

Marking Guidelines

Criteria	Marks
States the amplitude and period	2
Substantial progress eg states one of amplitude or period	1

Answer

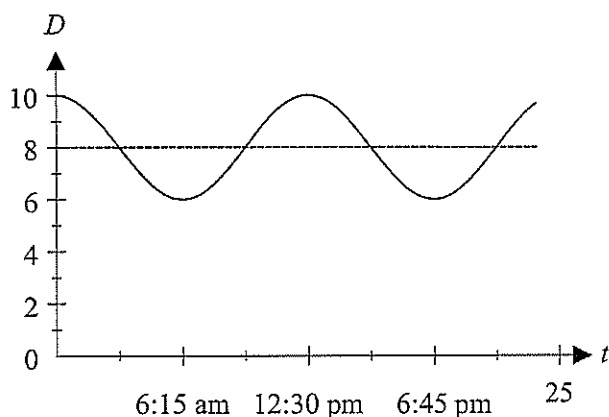
$$D = 8 + 2\cos\left(\frac{4\pi}{25}t\right) \quad \text{Amplitude 2 metres} \quad \text{Period } 2\pi \div \left(\frac{4\pi}{25}\right) = 12.5 \text{ hours}$$

b. Outcomes assessed: MA12-5

Marking Guidelines

Criteria	Marks
Sketches graph showing the detail required	2
Substantial progress eg sketches a cosine graph centred on $D = 8$ but some detail missing	1

Answer



Question 27

a. Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Calculates p correct to 4 decimal places	2
Substantial progress eg writes and simplifies a numerical expression for p^9	1

Answer

$$V = 20\,000p^t \quad t = 9, V = 32\,000 \Rightarrow p^9 = \frac{8}{5} \quad \therefore p = (1.6)^{\frac{1}{9}} \approx 1.0536$$

b. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Differentiates with respect to t then evaluates for $t = 9$ to find the rate to required accuracy	2
Substantial progress eg differentiates with respect to t	1

Answer

$$\frac{dV}{dt} = 20\,000 \ln p (p^t)$$

$$t = 9, \ln p = \frac{1}{9} \ln 1.6 \Rightarrow \frac{dV}{dt} = 20\,000 \left(\frac{1}{9} \ln 1.6\right) \times \frac{8}{5} \approx 1670$$

Rate of increase is \$1670 per year (nearest \$10 per year)

Question 28

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Describes nature and strength of the linear relationship	1

Answer

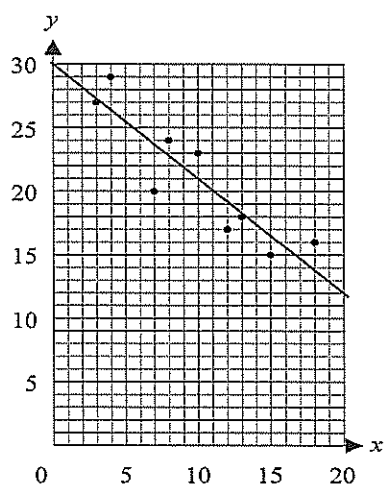
The scatter plot suggests a strong, negative linear relationship

b. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Sketches graph and states its equation	3
Substantial progress eg. sketches the line, giving equation with one of gradient, y intercept correct	2
Some progress eg sketches line	1

Answer



Line has equation $y = -0.9x + 30$

c. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Estimates the mark for J from the graph or the equation	1

Answer

$$y = -0.9x + 30 \quad x = 11 \Rightarrow y = 20.1$$

Hence using the equation, or using the graph, estimated mark for J is 20

Question 29

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Generalises expressions for P_1, P_2, P_3 to find expression P_n then uses sum of GP to simplify	3
Substantial progress eg process correct, but one error made, or simplification not completed	2
Some progress eg finds expressions for P_1, P_2, P_3	1

Answer

$$P_n = 1.1 \times P_{n-1} - 50$$

$$P_1 = 1.1 \times P_0 - 50$$

$$P_2 = 1.1 \times P_1 - 50 = 1.1 \times (1.1 \times P_0 - 50) - 50 = (1.1)^2 P_0 - 50(1.1 + 1)$$

$$P_3 = 1.1 \times P_2 - 50 = 1.1 \times \left\{ (1.1)^2 P_0 - 50(1.1 + 1) \right\} - 50 = (1.1)^3 P_0 - 50 \left\{ (1.1)^2 + 1.1 + 1 \right\}$$

Generalising: $P_n = (1.1)^n P_0 - 50 \left\{ 1 + 1.1 + (1.1)^2 + \dots + (1.1)^{n-1} \right\}$ where $\{ \dots \}$ is the sum of n terms of a GP with $a = 1, r = 1.1$

$$\begin{aligned} \text{Given } P_0 = 600, \quad P_n &= (1.1)^n \times 600 - 50 \frac{(1.1)^n - 1}{1.1 - 1} \quad \therefore P_n = 100 \times (1.1)^n + 500 \\ &= (1.1)^n \times 600 - 500 \left\{ (1.1)^n - 1 \right\} \end{aligned}$$

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Solves $P_n - P_{n-1} > 100$ to find the required year	2
Substantial progress eg correct procedure but one error in calculation	1

Answer

Require least integer value of n such that $P_n - P_{n-1} > 100$

$$100 \left\{ (1.1)^n - (1.1)^{n-1} \right\} > 100$$

$$(1.1)^{n-1} \{ 1.1 - 1 \} > 1$$

$$(1.1)^{n-1} > 10$$

$$(n-1) \log_{10}(1.1) > 1$$

$$n-1 > \frac{1}{\log_{10}(1.1)} \approx 24.16$$

$$n-1 = 25, 26, 27, \dots$$

Required year is 2026.

Question 30

a. Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Uses integration and initial conditions to obtain required result	2
Substantial progress eg integrates with respect to t	1

Answer

$$\begin{aligned} \frac{dx}{dt} &= 3(t+1)^{\frac{1}{2}} - 9 & t=0, x=0 \Rightarrow c=-2 \\ x &= 2(t+1)^{\frac{3}{2}} - 9t + c & \therefore x = 2(t+1)\sqrt{t+1} - 9t - 2 \end{aligned}$$

b. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Finds time and position when particle at rest	2
Substantial progress eg finds t when $v = 0$	1

Answer

$$\begin{aligned} v=0 &\Rightarrow \sqrt{t+1}=3 & \text{Also when } t=0, v=-3 \\ t &= 8, x = -20 \end{aligned}$$

Particle is at rest at time 8 seconds when 20 m from O in initial direction of travel.

c. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Finds distance travelled	2
Substantial progress eg finds x when $t = 24$	1

Answer

$$t = 24 \Rightarrow x = 32 \quad \text{Particle travels from } x=0 \text{ to } x=-20 \text{ then to } x=32 \text{ in first 24 seconds.}$$

Hence distance travelled is $20 + 20 + 32 = 72$ m

Question 31

a. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Writes and solves simultaneous equations for a and d	3
Substantial progress eg writes correct simultaneous equations but makes one error in solution	2
Some progress eg writes one of the equations correctly	1

Answer

$$\begin{array}{llll}
 248 = 64a + d & (1) & (2) - (1) \Rightarrow 224 = 448a & d = 248 - 64 \times \frac{1}{2} \\
 472 = 512a + d & (2) & \therefore a = \frac{1}{2} & d = 216
 \end{array}$$

b. Outcomes assessed: MA 11-1

Marking Guidelines

Criteria	Marks
Uses $S = TC$, substituting for a and d , to obtain required result	1

Answer

$$S = TC = \frac{120}{x} \left(\frac{x^3}{2} + 216 \right) \quad \therefore S = 60x^2 + \frac{25\,920}{x}$$

c. Outcomes assessed: MA 12-6

Marking Guidelines

Criteria	Marks
Uses differentiation to find the stationary point, verifying a minimum, to obtain minimum cost	3
Substantial progress eg correct procedure but one error or omission	2
Some progress eg differentiates correctly	1

Answer

$$\begin{array}{llll}
 \frac{dS}{dx} = 120x - \frac{25\,920}{x^2} & \frac{dS}{dx} = 0 \Rightarrow x = 6 & \text{Sign of } \frac{dS}{dx} & \begin{array}{c} - \quad 0 \quad + \\ | \quad | \quad | \\ 0 \quad 6 \quad x \end{array} \\
 = \frac{120}{x^2} (x^3 - 216) & S = 6480 & &
 \end{array}$$

Hence minimum cost is \$6480

Question 32

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Verifies both conditions are satisfied for $f(x)$ to be a probability density function	2
Substantial progress eg. finds and evaluates the anti-derivative	1

Answer

$$f(x) = \frac{2}{x^2}, \quad 1 \leq x \leq 2 \quad \int_1^2 f(x) dx = \left[-\frac{2}{x} \right]_1^2$$

$$\therefore f(x) > 0 \quad = -1 + 2 \quad \text{Hence } f(x) \text{ is a probability density function.}$$

$$= 1$$

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses integration to find the distribution function	2
Substantial progress eg correct procedure but one error or omission	1

Answer

$$F(x) = \int_1^x \frac{2}{x^2} dx$$

$$= \left[-\frac{2}{x} \right]_1^x \quad \therefore F(x) = \begin{cases} 0 & , x < 1 \\ 2\left(1 - \frac{1}{x}\right) & , 1 \leq x \leq 2 \\ 1 & , x > 2 \end{cases}$$

$$= -\frac{2}{x} + 2$$

c. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Finds the reciprocals of the quartiles; verifies a common difference between consecutive terms	3
Substantial progress eg correct procedure but an error in execution or AP not verified	2
Some progress eg. finds one of the quartiles or its reciprocal correctly	1

Answer

$$F(x) = \frac{1}{4} \Rightarrow 2\left(1 - \frac{1}{x}\right) = \frac{1}{4} \quad F(x) = \frac{1}{2} \Rightarrow 2\left(1 - \frac{1}{x}\right) = \frac{1}{2} \quad F(x) = \frac{3}{4} \Rightarrow 2\left(1 - \frac{1}{x}\right) = \frac{3}{4}$$

$$\frac{1}{x} = \frac{7}{8} \quad \frac{1}{x} = \frac{3}{4} \quad \frac{1}{x} = \frac{5}{8}$$

$$\therefore \frac{1}{Q_1} = \frac{7}{8} \quad \therefore \frac{1}{Q_2} = \frac{3}{4} \quad \therefore \frac{1}{Q_3} = \frac{5}{8}$$

$$\therefore \frac{1}{Q_2} - \frac{1}{Q_1} = -\frac{1}{8} \text{ and } \frac{1}{Q_3} - \frac{1}{Q_2} = -\frac{1}{8} \quad \therefore \frac{1}{Q_1}, \frac{1}{Q_2}, \frac{1}{Q_3} \text{ are consecutive terms in an AP.}$$

Question 33

a. Outcomes assessed: MA12-1

Marking Guidelines

Criteria	Marks
States a correct sequence of transformations giving appropriate detail	2
Substantial progress eg identifies one or two correct transformations	1

Answer

A vertical dilation by a factor 8, then reflections in the x and y axes, followed by a vertical translation upwards by 6 units.

b. Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Finds the required equation by equating expressions for y then solves exactly as a quadratic in e^x	3
Substantial progress eg finds equation and values of e^x	2
Some progress eg obtains the required equation	1

Answer

Curves intersect where $e^x = 6 - 8e^{-x}$
 $e^{2x} = 6e^x - 8$

$$\begin{aligned} e^{2x} - 6e^x + 8 &= 0 & \therefore e^x = 2 \text{ or } e^x = 4 \\ (e^x - 2)(e^x - 4) &= 0 & \therefore x_1 = \ln 2, \quad x_2 = \ln 4 \end{aligned}$$

c. Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Writes and evaluates the appropriate definite integral giving answer in simplest exact form	3
Substantial progress eg. writes definite integral and finds anti-derivative with partial evaluation	2
Some progress eg writes definite integral and finds anti-derivative	1

Answer

$$\begin{aligned} \text{Area is given by } \int_{\ln 2}^{\ln 4} (6 - 8e^{-x} - e^x) dx &= \left[6x + 8e^{-x} - e^x \right]_{\ln 2}^{\ln 4} \\ &= 6(\ln 4 - \ln 2) + 8(e^{-\ln 4} - e^{-\ln 2}) - (e^{\ln 4} - e^{\ln 2}) \\ &= 6\ln 2 + 8\left(\frac{1}{4} - \frac{1}{2}\right) - (4 - 2) \\ &= 6\ln 2 - 4 \end{aligned}$$

Hence area is $(6\ln 2 - 4)$ sq. units.

Independent HSC Trial Examinations 2022

Mathematics Advanced

Mapping Grid

Question	Marks	Content	Syllabus Outcomes	Targeted Performance Bands
1	1	Trigonometry and measure of angles	MA11-3	2-3
2	1	Working with functions	MA11-2	2-3
3	1	Probability and discrete probability distributions	MA11-7	3-4
4	1	Working with functions	MA11-2	3-4
5	1	Graphing techniques	MA12-1	3-4
6	1	Differential calculus	MA12-6	4-5
7	1	Random variables	MA12-8	4-5
8	1	Integral calculus	MA12-7	4-5
9	1	Modelling financial situations	MA12-2	5-6
10	1	Modelling financial situations	MA12-4	5-6
11	2	Working with functions	MA11-2	2-3
12	2	Trigonometric functions and identities	MA11-3	2-3
13	2	Probability and discrete probability distributions	MA11-7	2-3
14	3	Working with functions	MA11-1	2-3
15	3	Introduction to differentiation	MA11-5	2-3
16 a	2	Trigonometry and measure of angles	MA11-3	2-3
b	2	Trigonometry and measure of angles	MA11-3	4-5
17 a	1	Probability and discrete probability distributions	MA11-7	2-3
b	1	Probability and discrete probability distributions	MA11-7	3-4
c	2	Probability and discrete probability distributions	MA11-7	4-5
18	2	Logarithms and exponentials	MA11-6	2-3
19	2	Differential calculus	MA12-6	3-4
20	2	Applications of differentiation	MA12-6	3-4
21 a	2	Random variables	MA12-8	3-4
b	2	Random variables	MA12-8	3-4
22	3	Integral calculus	MA12-7	4-5
23	3	Integral calculus	MA12-7	5-6
24	4	Applications of differentiation	MA12-6	5-6
25 a	2	Modelling financial situations	MA12-2	4-5
b	2	Modelling financial situations	MA12-2	4-5
26 a	2	Trigonometric functions and graphs	MA12-5	3-4
b	2	Trigonometric functions and graphs	MA12-5	4-5
27 a	2	Logarithms and exponentials	MA11-6	3-4
b	2	Differential calculus	MA12-6	4-5
28 a	1	Descriptive statistics and bivariate data analysis	MA12-8	2-3
b	3	Working with functions	MA11-1	3-4
c	1	Working with functions	MA11-1	3-4
29 a	3	Modelling financial situations	MA12-2	5-6
b	2	Modelling financial situations	MA12-2	5-6
30 a	2	Integral calculus	MA12-7	5-6
b	2	Applications of differentiation	MA12-6	3-4
c	2	Working with functions	MA11-1	4-5
31 a	3	Working with functions	MA11-1	2-3
b	1	Working with functions	MA11-1	2-3
c	3	Applications of differentiation	MA12-6	5-6
32 a	2	Random variables	MA12-8	3-4
b	2	Random variables	MA12-8	3-4
c	3	Random variables	MA12-8	5-6
33 a	2	Graphing techniques	MA12-1	4-5
b	3	Logarithms and exponentials	MA11-6	4-5
c	3	Integral calculus	MA12-7	5-6